

Basic Database Operations



**Skills
Network**

Estimated Time Needed: 45 min

In this lab, you will create a database, import data to populate a table, and perform basic database operations in the MySQL database service using the phpMyAdmin graphical user interface (GUI) tool.

Objectives

After completing this lab, you will be able to use phpMyAdmin with MySQL to:

- Create and populate a database and table
- Execute Structured Query Language (SQL) commands to perform basic database operations
- Retrieve data from tables using simple SELECT statements
- Learn how to filter the data output using WHERE statements
- Add and remove data from a table using INSERT and DELETE statements
- Remove a table using DROP statements

Software used in this lab

Software used in this lab



You will use MySQL to complete this lab. MySQL is a free, open source relational database system that offers a command line interface (MySQL) and a third-party web interface (phpMyAdmin) to efficiently store, manipulate, and retrieve data.

MySQL is a service available on Skills Network Labs (SN Labs) Cloud IDE, a virtual lab environment used in this course. SN Labs Cloud IDE is great way to do projects without downloading, installing, configuring, and integrating software on your own computer.

Two Components of the SN Labs Cloud IDE

1. The instructions that you will follow to complete this lab are displayed on the left side of the screen.
2. The area on the right side of the screen is where you will use the menus, terminals, and tools to complete the lab exercises.

Skills Network Labs

Instructions

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Step 1 of 13 ▶

Basic Database Operations



1

View Lab Instructions in this area

Estimated Time Needed: 35 min

In this lab, you will create a database, import data to populate a table, and perform basic database operations in the MySQL database service using the phpMyAdmin graphical user interface (GUI) tool.

Continue

Lab

IBMCLOUD OpenShift Console

File Edit Selection View Go

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Dataset used in this lab

Dataset used in this Lab

The dataset used in this lab is single Open Document Spreadsheet file (.ods) that contains 10 (ten) rows and 6 (six) columns of fictitious customer demographics and insurance policy data. To complete the exercises in this lab, you will be instructed to save and upload the dataset [Insurance-info.ods](#).

PolicyID	ExpiryDate	State	Region	ConstructionType	Business
10042	1/2/2022	NY	East	Masonry	Apartment
10081	1/21/2021	NY	East	Frame	Farming
10075	6/25/2022	OH	Central	Masonry	Medical
10123	1/25/2021	WI	Midwest	Masonry	Service
10019	3/23/2021	OH	Central	Frame	Office
10132	1/19/2021	NJ	East	Masonry	Organization
10466	4/3/2021	NJ	East	Frame	Hospitality
11088	1/5/2021	NY	East	Frame	Retail
12091	1/31/2022	IL	Midwest	Masonry	Apartment
14202	2/4/2021	NY	East	Masonry	Construction

Lab Exercises

Lab Exercises

Through a series of exercises, you will perform basic operations in a relational database system:

- Exercise 1: Create a database
- Exercise 2: Import a table and data
- Exercise 3: Select data from a database
- Exercise 4: Filter data output
- Exercise 5: Insert a new row to a table
- Exercise 6: Remove a row from a table
- Exercise 7: Remove a table from a database

Exercise 1: Create a database

Exercise 1: Create a database

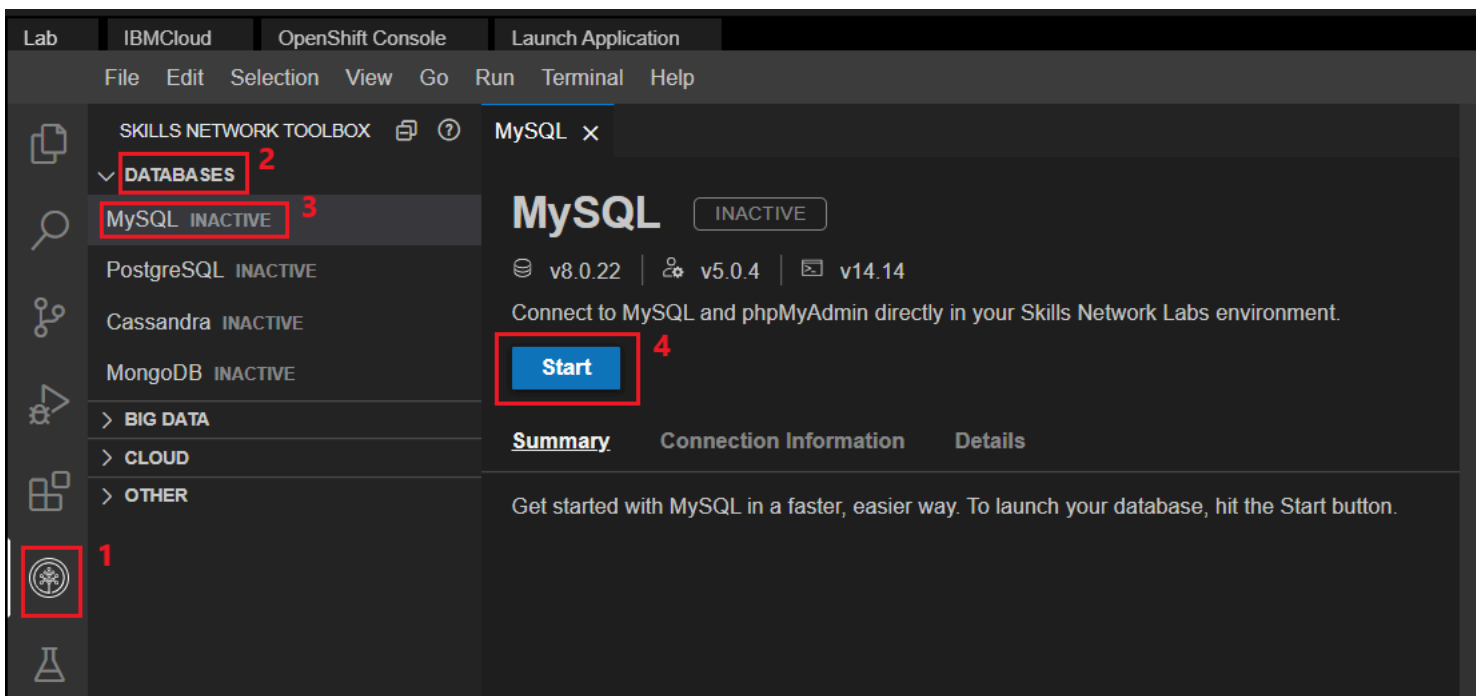
Task A: Start MySQL and phpMyAdmin GUI

Step 1: In the navigation toolbar on the left side of the SN Labs Cloud IDE (also known as the virtual lab environment), select the Skills Network Toolbox icon.

Step 2: In the Toolbox options, select "Databases".

Step 3: In the Databases sub-menu, select "MySQL".

Step 4: Select the "Start" button to connect to MySQL. It may take up to a couple of minutes to start the MySQL database server.



Step 5: When the status next to "MySQL" is "Active", select the "phpMyAdmin" button to launch the GUI.

Lab | IBMCloud | OpenShift Console | Launch Application

File Edit Selection View Go Run Terminal Help

SKILLS NETWORK TOOLBOX MySQL x

DATABASES

- MySQL **ACTIVE**
- PostgreSQL INACTIVE
- Cassandra INACTIVE
- MongoDB INACTIVE

MySQL **ACTIVE**

v8.0.22 | v5.0.4 | v14.14

Connect to MySQL and phpMyAdmin directly in your Skills Network Labs environment.

Stop

Summary Connection Information Details

Your database and phpMyAdmin server are now ready to use and available with the following login credentials. For more details on how to navigate MySQL, please check out the Details section.

Username: jennifer1236

Password: MjMyMzQtamVubmlm

You can manage MySQL via:

5 **phpMyAdmin**

Or to interact with the database in the terminal, select one of these options:

MySQL CLI **New Terminal**

Go Live

Task B: Create a new database

A database is a data structure that stores organized information and is accessed electronically. A database consists of one or more tables that are made up of multiple rows and columns of data.

Step 1: Using the top menu of the phpMyAdmin screen, select the "Databases" tab.

Step 2: Type "Insurance" in the text field that currently displays "Database Name".

Step 3: Lastly, select the "Create" button.

Server: mysql:3306

Databases SQL Status User accounts More

Databases

Create database

Insurance utf8mb4_0900_ai_ci Create

Database	Collation	Master replication	Action
☐ information_schema	utf8_general_ci	✓ Replicated	Check privileges
☐ mysql	utf8mb4_0900_ai_ci	✓ Replicated	Check privileges
☐ performance_schema	utf8mb4_0900_ai_ci	✓ Replicated	Check privileges
☐ sys	utf8mb4_0900_ai_ci	✓ Replicated	Check privileges

Total: 4

☐ Check all With selected: Drop

Note: Enabling the database statistics here might cause heavy traffic between the web server and the MySQL server.

Enable statistics

Console

Go Live

Exercise 2: Import a table and data

Exercise 2: Import a table and data

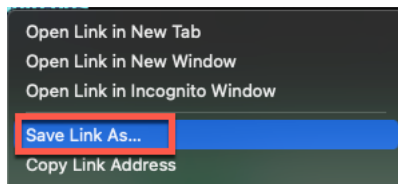
A table is a collection of related data that is stored in columns and rows like an Excel spreadsheet. The columns of the table, also known as "fields", are the vertical groupings (or categories) of data. The rows of the table, known as "records", are the horizontal groupings of unique data.

Task A: Add table and import data

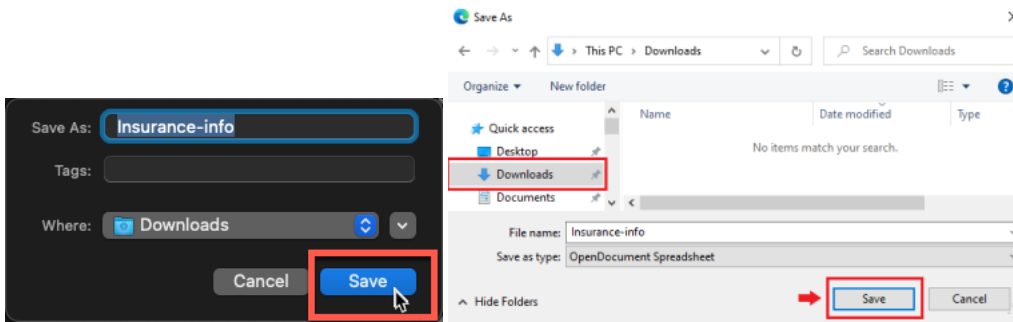
Step 1: Download the dataset to your local computer by **Right-Clicking** on the following link and choose **Save Link As...**:

[Insurance-info.ods](#)

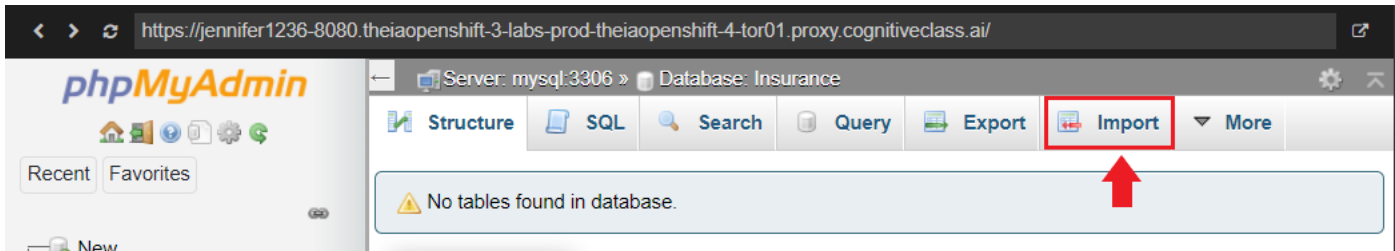
- You will be prompted for a location to download the file on your computer. The prompt may look a little different than the one below depending on your web browser.



- Select a suitable destination and **Save** the file on your device. The prompt may look a little different than the one below depending on your operating system.



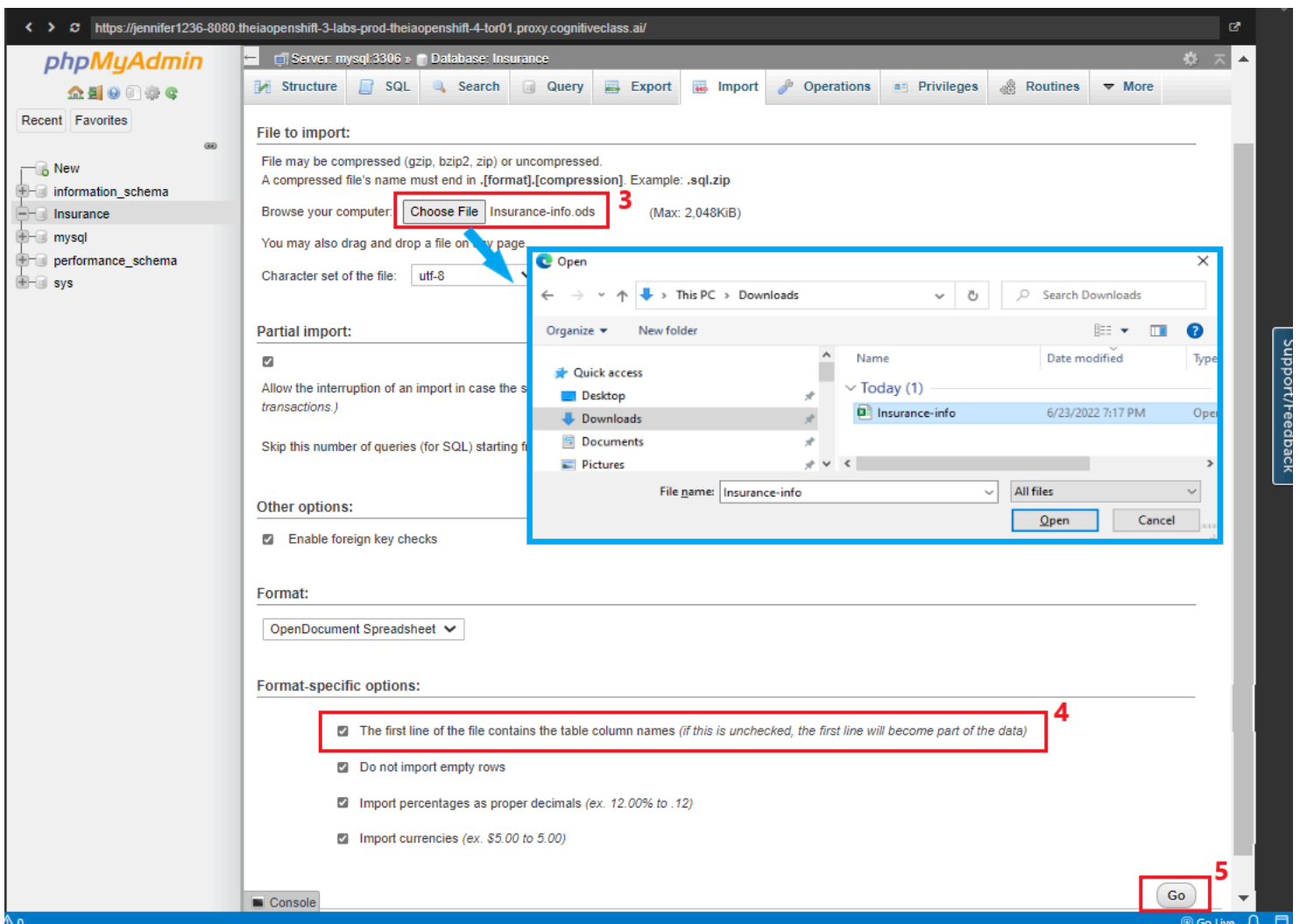
Step 2: In the phpMyAdmin screen, select the "Import" tab from the top menu.



Step 3: In the next screen, select "Choose File" to and the file "Insurance-info.ods" from your computer.

Step 4: Scroll down to "Format-specific options" and select the first checkbox to indicate: "The first line of file contains table column names."

Step 5: Select the "Go" button.



When these steps are completed successfully, you will see the following screen that confirms:

1. The table named Policy has been created.
2. 10 rows and 6 columns of data were added to the table.

Exercise 3: Select data from a database

Exercise 3: Select data from a database

Structured Query Language (SQL) Statements, also known as "commands", are used to communicate with a database to retrieve data or modify its structure and its contents.

The SELECT statement is a database query that is used to request data from the relevant rows and columns in a table. The output of the query is stored and displayed in a table known as the SQL result-set.

Syntax of a SELECT Statement

```
SELECT column1, column2
FROM table_name;
```

- The first line selects the field/column names of the data to be displayed.
- The second line indicates the name of the table where the selected data is stored.

Task A: Review the Source Code

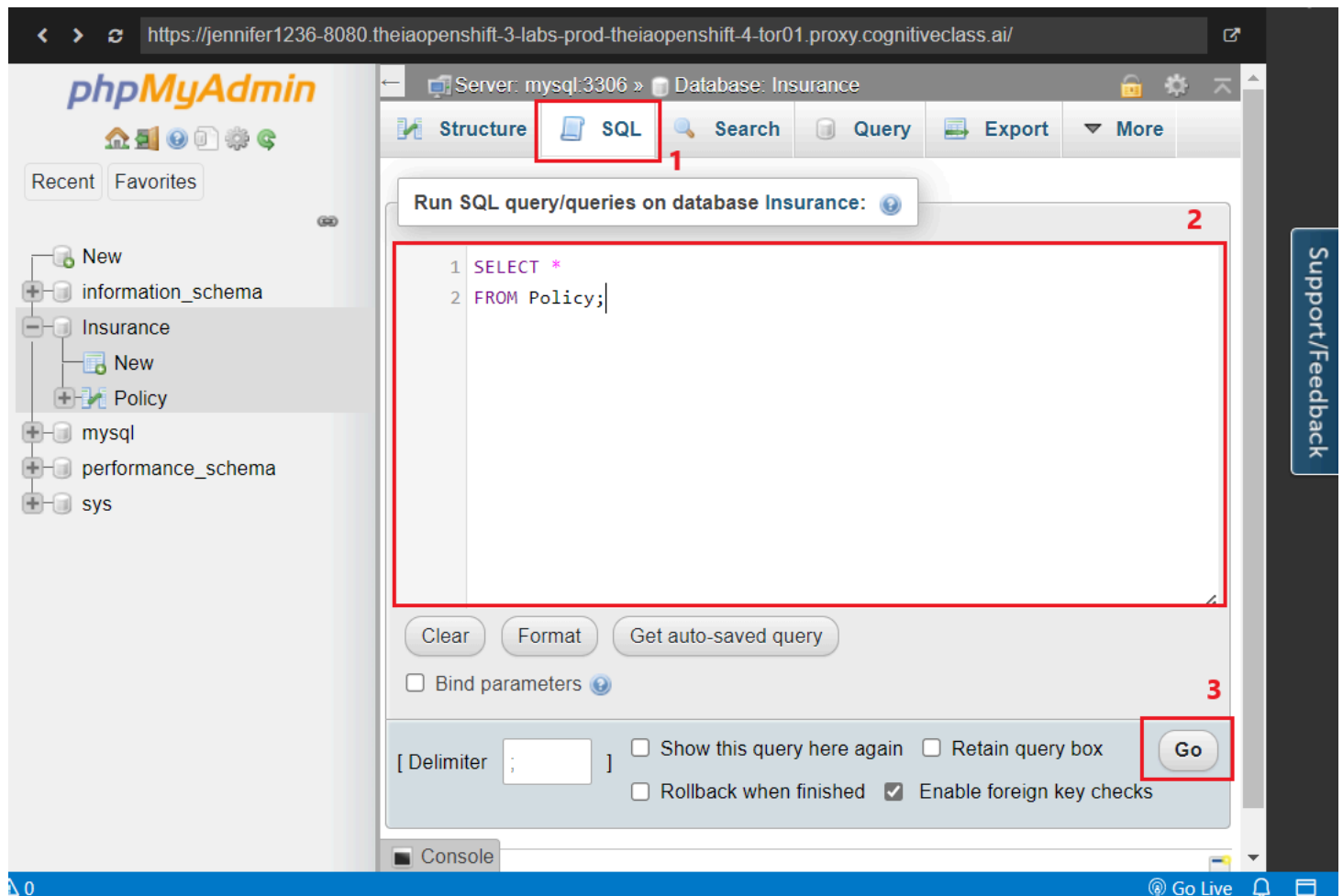
Step 1: In the phpMyAdmin screen, select the "SQL" tab from the top menu to access the query box.

Step 2: Copy the code block below and paste it into the query box.

```
SELECT *
```

- The asterisk (*) in the first line of this SELECT statement is a wildcard character used to indicate that data from all columns should be returned in the query output.

Step 3: Select the "Go" button.



Step 4: Review the query output. When these steps are completed correctly, the query output will display all 6 columns and 10 rows of data from the Policy table.

Server: mysql:3306 » Database: Insurance » Table: Policy

Browse Structure SQL Search Insert Export Import More

Show query box

⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

✓ Showing rows 0 - 9 (10 total, Query took 0.0005 seconds.)

`SELECT * FROM Policy`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

+ Options

PolicyID	ExpiryDate	State	Region	ConstructionType	Business
10042	1/2/2022	NY	East	Masonry	Apartment
10081	1/21/2021	NY	East	Frame	Farming
10075	6/25/2022	OH	Central	Masonry	Medical
10123	1/25/2021	WI	Midwest	Masonry	Service
10019	3/23/2021	OH	Central	Frame	Office
10132	1/19/2021	NJ	East	Masonry	Organization
10466	4/3/2021	NJ	East	Frame	Hospitality
11088	1/5/2021	NY	East	Frame	Retail
12091	1/31/2022	IL	Midwest	Masonry	Apartment
14202	2/4/2021	NY	East	Masonry	Construction

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Console

Task B: Write the SQL Statement

Step 1: Select the "SQL" tab from the top menu to access the query box.

Server: mysql:3306 » Database: Insurance » Table: Policy

Browse Structure SQL Search Insert More

Step 2: Use the "Clear" button to delete the previous SQL Statement.

Step 3: In the query box, write a SELECT statement to display the columns PolicyID and ConstructionType from the table named Policy.

Step 4: Select the "Go" button.

Run SQL query/queries on table Insurance.Policy:

3 Columns

PolicyID
ExpiryDate
State
Region
ConstructionType
Business

1

SELECT * SELECT INSERT UPDATE

DELETE Clear Format

Get auto-saved query

☐ Bind parameters

Console ;] ☐ Show this query here again ☐ Retain query box

Go

Step 5: Review the query output. When these steps are completed correctly, the query output will display 2 columns and 10 rows of data from the Policy table.

PolicyID	ConstructionType
10042	Masonry
10081	Frame
10075	Masonry
10123	Masonry
10019	Frame
10132	Masonry
10466	Frame
11088	Frame
12091	Masonry
14202	Masonry

► [Click here for Hint](#)

► [Click here for Solution](#)

Exercise 4: Filter data output

Exercise 4: Filter data output

The WHERE statement defines a condition used to filter records and limit the amount of data returned in the query output.

Syntax of a WHERE Statement

```
SELECT column1, column2, ...
FROM table_name
WHERE condition;
```

- The first line selects the field/column names of the data to be displayed.
- The second line indicates the name of the table where the selected data is stored.
- The third line defines the condition to test and limit the data in the query output.

Task A: Review the Source Code

Step 1: Select the "SQL" tab from the top menu to access the query box.

Step 2: Use the "Clear" button to delete the previous SQL Statement.

Step 3: Copy and paste the code block below to the query box to filter and display the records from the Policy table that have Masonry construction.

```
SELECT *
FROM Policy
WHERE ConstructionType = "Masonry";
```

Step 4: Select the "Go" button.

Step 5: Review the query output.

- When these steps are completed correctly, the query output will display 6 columns and 6 rows of data from the Policy table.

The screenshot shows the phpMyAdmin interface with the following details:

- URL:** <https://jennifer1236-8080.theiaopenshift-3-labs-prod-theiaopenshift-4-tor01.proxy.cognitiveclass.ai/>
- Navigation Panel:** Shows database structure with 'Insurance' selected and 'Policy' table highlighted.
- Top Menu:** Browse, Structure, SQL (selected), Search, Insert, Export, Import, More.
- Query Box:** Contains the SQL query: `SELECT * FROM Policy WHERE ConstructionType = "Masonry";`
- Message Bar:** Shows a success message: "Showing rows 0 - 5 (6 total, Query took 0.0004 seconds.)"
- Table Data:**

PolicyID	ExpiryDate	State	Region	ConstructionType	Business
10042	1/2/2022	NY	East	Masonry	Apartment
10075	6/25/2022	OH	Central	Masonry	Medical
10123	1/25/2021	WI	Midwest	Masonry	Service
10132	1/19/2021	NJ	East	Masonry	Organization
12091	1/31/2022	IL	Midwest	Masonry	Apartment
14202	2/4/2021	NY	East	Masonry	Construction
- Query results operations:** Print, Copy to clipboard, Export, Display chart, Create view.

Task B: Write the SQL Code

Step 1: Select the "SQL" tab from the top menu to access the query box.

Step 2: Use the "Clear" button to delete the previous SQL Statement.

Step 3: Using the Source Code in Task A as an example, write a query to display contents of the Policy table where the Region is East.

Step 4: Select the "Go" button.

Step 5: Review the query output. When these steps are completed correctly, the query output will display 6 columns and 6 rows of data from the Policy table.

PolicyID	ExpiryDate	State	Region	ConstructionType	Business
10042	1/2/2022	NY	East	Masonry	Apartment
10081	1/21/2021	NY	East	Frame	Farming
10132	1/19/2021	NJ	East	Masonry	Organization
10466	4/3/2021	NJ	East	Frame	Hospitality
11088	1/5/2021	NY	East	Frame	Retail
14202	2/4/2021	NY	East	Masonry	Construction

► Click here for Hint

► Click here for Solution

Exercise 5: Insert a new row to a table

Exercise 5: Insert a new row to a table

To insert a new record/row to a table, use the SQL statement `INSERT INTO`.

Syntax of an `INSERT INTO` Statement

```
INSERT INTO table_name
VALUES (value1, value2, value3, ...);
```

- The first line indicates the name of table where the new record will be added.
- The second line assigns values in the same order as the columns in the table.

Task A: Review the Source Code

Step 1: Select the "SQL" tab from the top menu to access the query box.

Step 2: Use the "Clear" button to delete the previous SQL Statement.

Step 3: Copy and paste the code block below to the query box to insert a row into the Policy table.

```
INSERT INTO Policy
values (100329, '4/10/2021', 'MN', 'Midwest', 'Frame', 'Hospitality');
```

The values, written in list format, for the source code above are as follows:

- PolicyID = 100329
- ExpiryDate = 4/10/2021
- State = MN
- Region = Midwest
- ConstructionType = Frame
- Business = Hospitality

Step 4: Select the "Go" button.

When these steps are done correctly, you will see the following message:

✓ 1 row inserted. (Query took 0.0036 seconds.)

```
INSERT INTO Policy values (100329, '4/10/2021', 'MN', 'Midwest', 'Frame', 'Hospitality')
```

Step 5: Review the updated table. Return to the query box and run the SQL Statement: `SELECT * from Policy` to see there are now 11 records in the table.

The screenshot shows the phpMyAdmin interface for a MySQL database named 'Insurance'. The 'Policy' table is selected, and the 'SQL' tab is active. The query box contains the statement `SELECT * FROM `Policy` WHERE 1`. A message indicates that the current selection does not contain a unique column, so grid edit, checkbox, Edit, Copy, and Delete features are not available. The table displays 11 rows, with the last row (PolicyID 100329) highlighted in green and marked with a red arrow. The table structure is as follows:

PolicyID	ExpiryDate	State	Region	ConstructionType	Business
10042	1/2/2022	NY	East	Masonry	Apartment
10081	1/21/2021	NY	East	Frame	Farming
10075	6/25/2022	OH	Central	Masonry	Medical
10123	1/25/2021	WI	Midwest	Masonry	Service
10019	3/23/2021	OH	Central	Frame	Office
10132	1/19/2021	NJ	East	Masonry	Organization
10466	4/3/2021	NJ	East	Frame	Hospitality
11088	1/5/2021	NY	East	Frame	Retail
12091	1/31/2022	IL	Midwest	Masonry	Apartment
14202	2/4/2021	NY	East	Masonry	Construction
100329	4/10/2021	MN	Midwest	Frame	Hospitality

Task B: Write the SQL Code

Step 1: Select the "SQL" tab from the top menu to access the query box.

Step 2: Use the "Clear" button to delete the previous SQL Statement.

Step 3: Using the same code syntax as above, write an INSERT INTO statement to add one row to the table Policy with the following values:

- PolicyID = 100500
- ExpiryDate = 5/5/2021
- State = NJ
- Region = East
- ConstructionType = Masonry
- Business = Office

Step 4: Select the "Go" button.

When these steps are done correctly, you will see the following message:

✓ 1 row inserted. (Query took 0.0036 seconds.)

Step 5: Review the updated table. Return to the query box and run the SQL Statement: `SELECT * from Policy` to see there are now 12 records in the table.

PolicyID	ExpiryDate	State	Region	ConstructionType	Business
10042	1/2/2022	NY	East	Masonry	Apartment
10081	1/21/2021	NY	East	Frame	Farming
10075	6/25/2022	OH	Central	Masonry	Medical
10123	1/25/2021	WI	Midwest	Masonry	Service
10019	3/23/2021	OH	Central	Frame	Office
10132	1/19/2021	NJ	East	Masonry	Organization
10466	4/3/2021	NJ	East	Frame	Hospitality
11088	1/5/2021	NY	East	Frame	Retail
12091	1/31/2022	IL	Midwest	Masonry	Apartment
14202	2/4/2021	NY	East	Masonry	Construction
100329	4/10/2021	MN	Midwest	Frame	Hospitality
100500	5/5/2021	NJ	East	Masonry	Office



► Click here for Hint

► Click here for Solution

Exercise 6: Remove a row from a table

Exercise 6: Remove a row from a table

The DELETE command is used to remove existing records from a table.

Syntax of a DELETE Statement

```
DELETE FROM table_name
WHERE condition;
```

- The first line indicates the name of the table that records will be deleted from.
- The second line defines the condition and deletes the record if the condition is true.

Task A: Review the Source Code

Step 1: Select the "SQL" tab from the top menu to access the query box.

Step 2: Use the "Clear" button to delete the previous SQL Statement.

Step 3: Copy and paste the code block below to delete the record for PolicyID 100329.

```
Delete From Policy
Where PolicyID=100329;
```

Step 4: Select the "Go" button.

- A pop-up will appear to confirm that you want to delete the row.

...ker-2-labs-prod-theiak8s-4-tor01.proxy.cognitiveclass.ai says

Do you really want to execute "Delete From Policy
Where PolicyID=100329;"?

OK

Cancel

- Select "OK" to proceed running the code and deleting the record.
- When these steps are done correctly, you will see the following message:

✓ 1 row affected. (Query took 0.0035 seconds.)

Delete From Policy Where PolicyID=100329

Step 5: Review the updated table. Return to the query box and run the SQL Statement: `SELECT * from Policy` to see there are now 11 records in the table.

The screenshot shows the phpMyAdmin interface for the 'Insurance' database. The 'Policy' table is selected, and the 'SQL' tab is active. A message indicates that 1 row was affected by a previous query. The SQL query box contains the statement: `SELECT * FROM `Policy` WHERE 1`. Below the query box, a table displays 11 rows of data. The table has columns: PolicyID, ExpiryDate, State, Region, ConstructionType, and Business. The rows are as follows:

PolicyID	ExpiryDate	State	Region	ConstructionType	Business
10042	1/2/2022	NY	East	Masonry	Apartment
10081	1/21/2021	NY	East	Frame	Farming
10075	6/25/2022	OH	Central	Masonry	Medical
10123	1/25/2021	WI	Midwest	Masonry	Service
10019	3/23/2021	OH	Central	Frame	Office
10132	1/19/2021	NJ	East	Masonry	Organization
10466	4/3/2021	NJ	East	Frame	Hospitality
11088	1/5/2021	NY	East	Frame	Retail
12091	1/31/2022	IL	Midwest	Masonry	Apartment
14202	2/4/2021	NY	East	Masonry	Construction
100500	5/5/2021	NJ	East	Masonry	Office

Task B: Write the SQL Code

Step 1: Select the "SQL" tab from the top menu to access the query box.

Step 2: Use the "Clear" button to delete the previous SQL Statement.

Step 3: Write a query to delete all rows that are in Midwest region.

Step 4: Select the "Go" button.

When these steps are done correctly, you will see the following message:

✓ 2 rows affected. (Query took 0.0033 seconds.)

Step 5: Review the updated table. Return to the query box and run the SQL Statement: `SELECT * from Policy` to see there are now 9 records in the table.

PolicyID	ExpiryDate	State	Region	ConstructionType	Business
10042	1/2/2022	NY	East	Masonry	Apartment
10081	1/21/2021	NY	East	Frame	Farming
10075	6/25/2022	OH	Central	Masonry	Medical
10019	3/23/2021	OH	Central	Frame	Office
10132	1/19/2021	NJ	East	Masonry	Organization
10466	4/3/2021	NJ	East	Frame	Hospitality
11088	1/5/2021	NY	East	Frame	Retail
14202	2/4/2021	NY	East	Masonry	Construction
100500	5/5/2021	NJ	East	Masonry	Office

► [Click here for Hint](#)

► [Click here for Solution](#)

Exercise 7: Remove a table from the database

Exercise 7: Remove a table from the database

To remove an existing table from the database, use the DROP command.

Syntax of a DROP Statement

```
DROP TABLE table_name;
```

- This line indicates the name of the table that will be deleted from the database.

Task A: Write the SQL Code

Step 1: Select the "SQL" tab from the top menu to access the query box.

Step 2: Use the "Clear" button to delete the previous SQL Statement.

Step 3: Copy and paste the code block below to delete the table Policy from the database.

```
DROP TABLE Policy;
```

Step 4: Select the "Go" button.

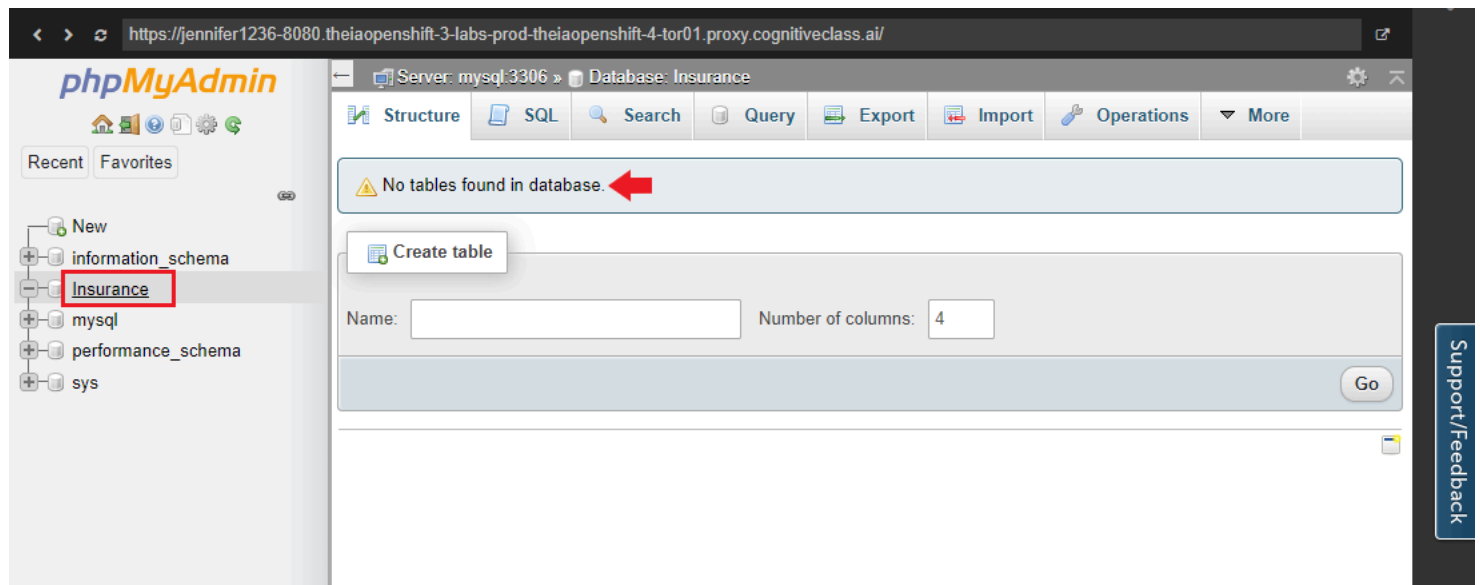
- A pop-up will appear to confirm that you want to delete the table.
- Select "OK" to proceed running the code and deleting the table.
- When these steps are done correctly, you will see the following message:

✔ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0100 seconds.)

```
DROP TABLE Policy
```

Step 5: Review the updated database.

- Select "Insurance" from the left-hand menu to confirm there are no tables in the Insurance database.



Congratulations!

Congratulations! You have completed the lab and you are ready for the next topic.

Author(s)

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